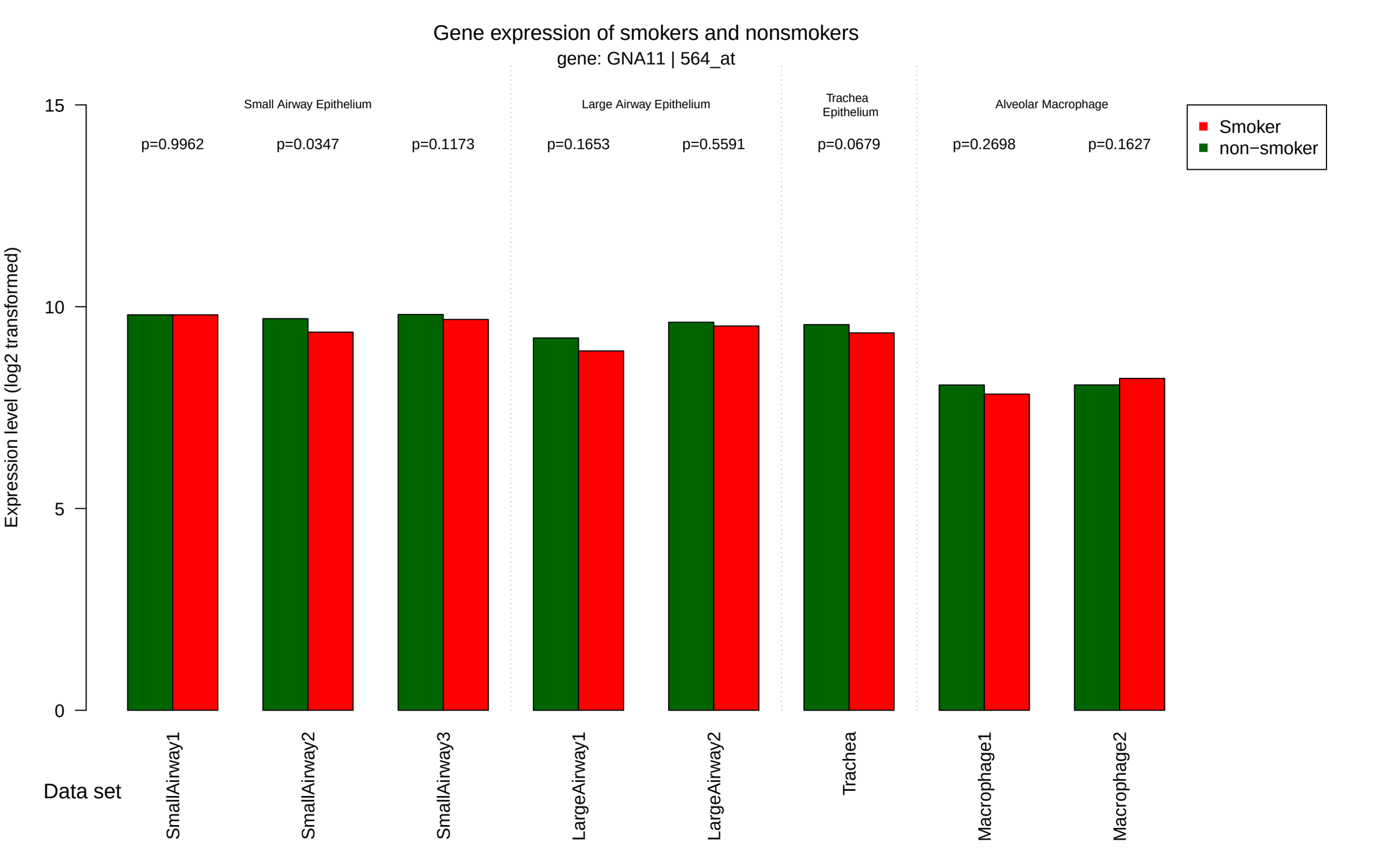
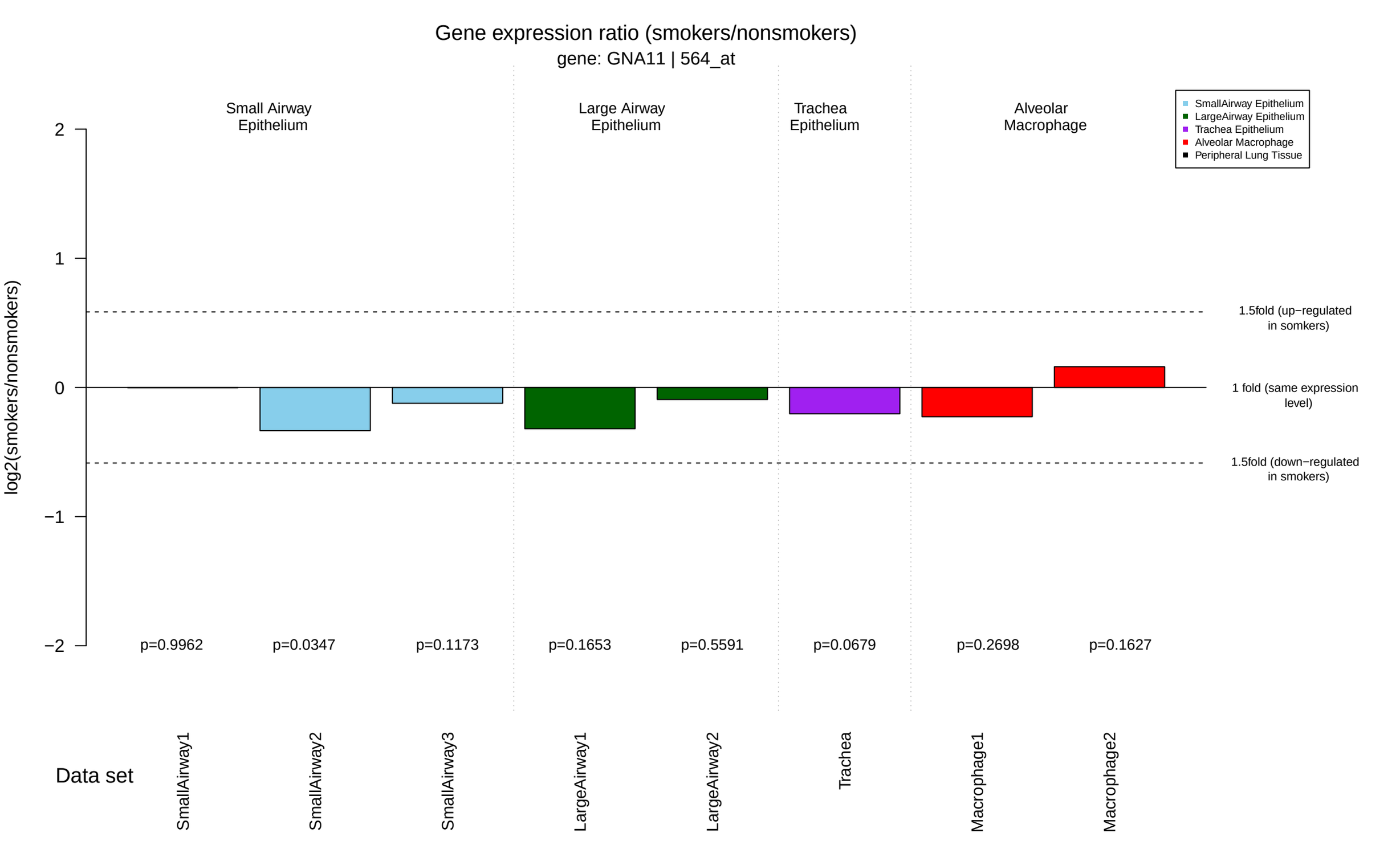
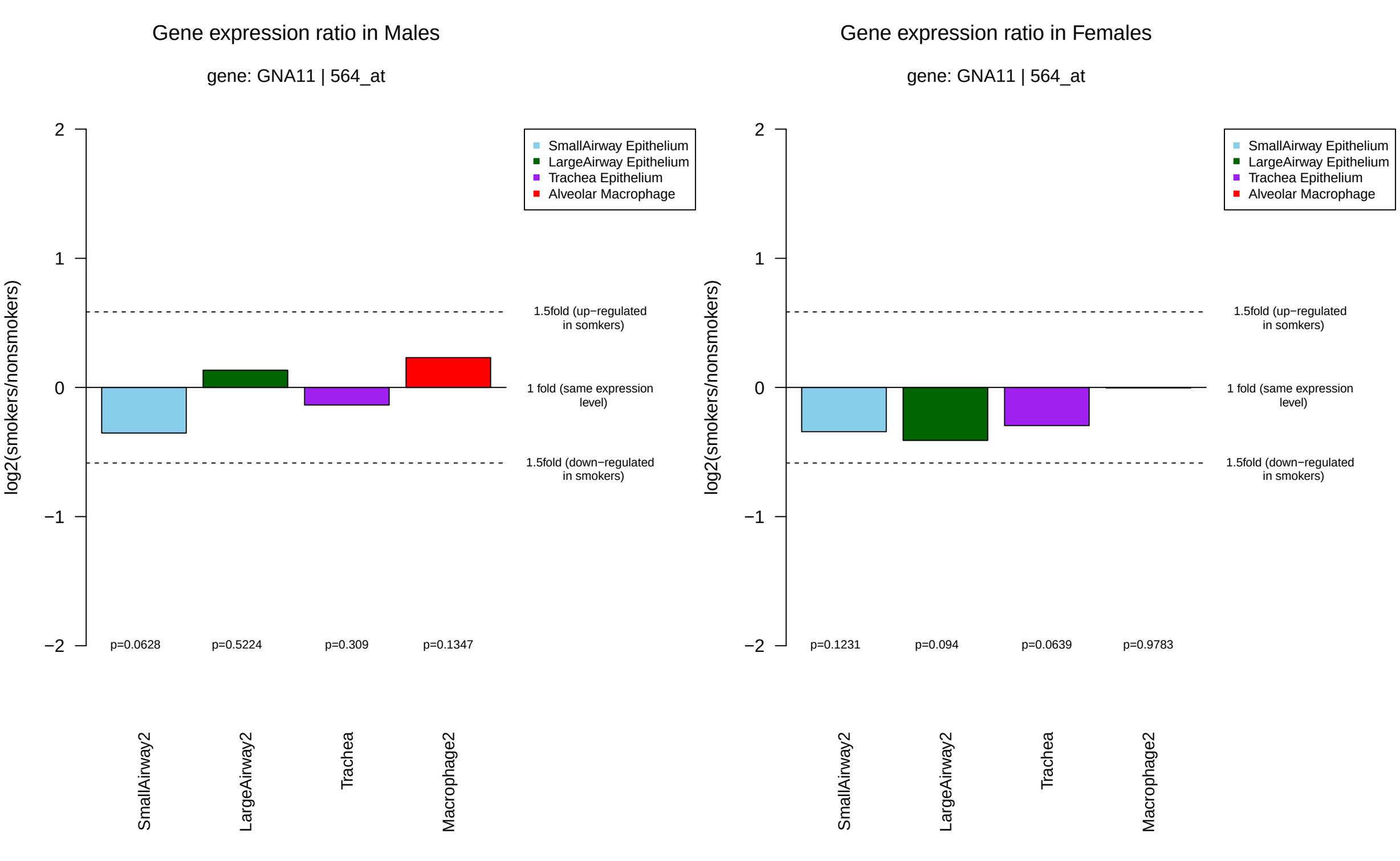
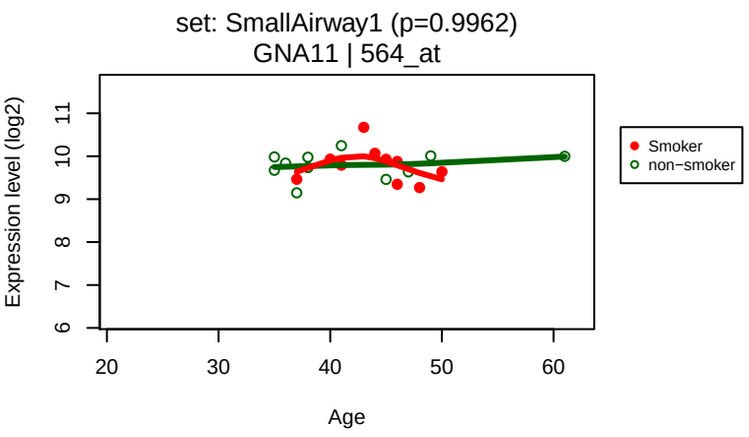
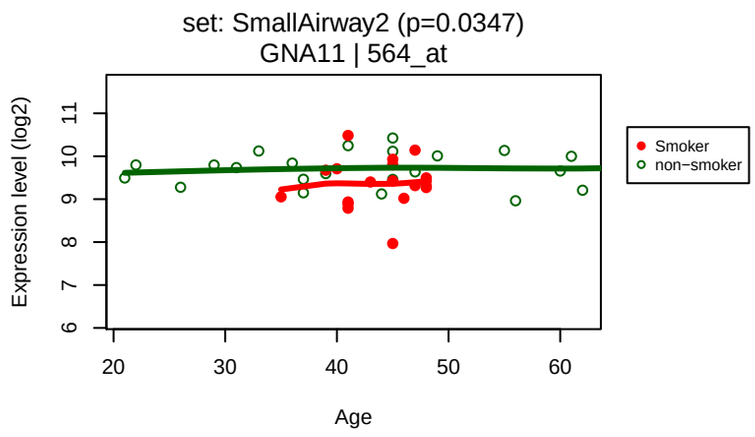
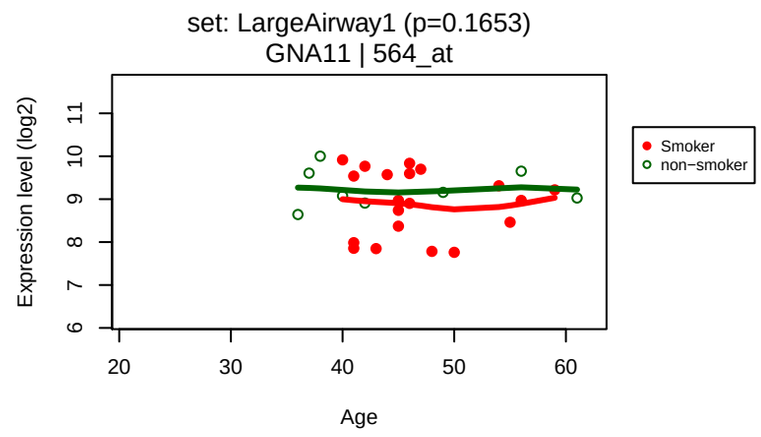

Gene expression of smokers and nonsmokers
gene: GNA11 | 564_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9962
p=0.0347
p=0.1173
p=0.1653
p=0.5591
p=0.0679
p=0.2698
p=0.1627
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

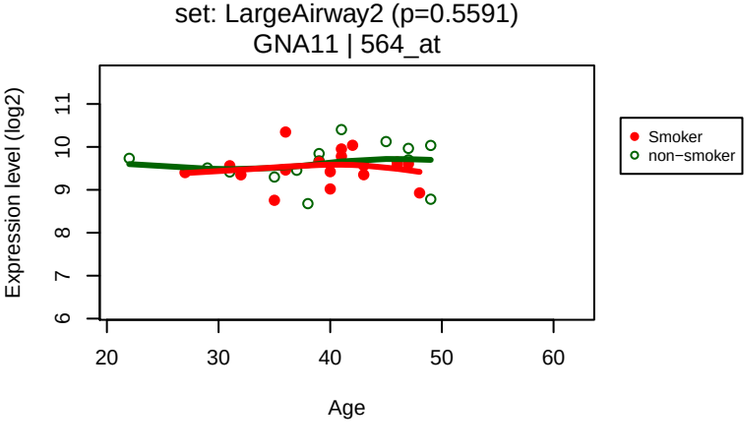

Gene expression ratio (smokers/nonsmokers)
gene: GNA11 | 564_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9962
p=0.0347
p=0.1173
p=0.1653
p=0.5591
p=0.0679
p=0.2698
p=0.1627
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

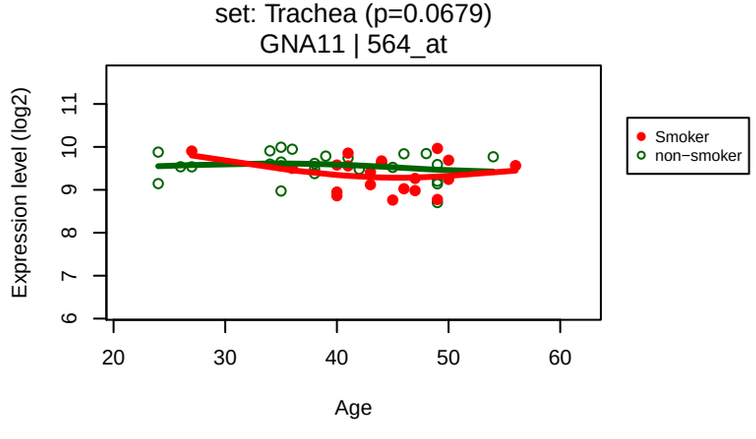

Gene expression ratio in Males
gene: GNA11 | 564_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0628
p=0.5224
p=0.309
p=0.1347
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNA11 | 564_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1231
p=0.094
p=0.0639
p=0.9783
SmallAirway2
LargeAirway2
Trachea
Macrophage2

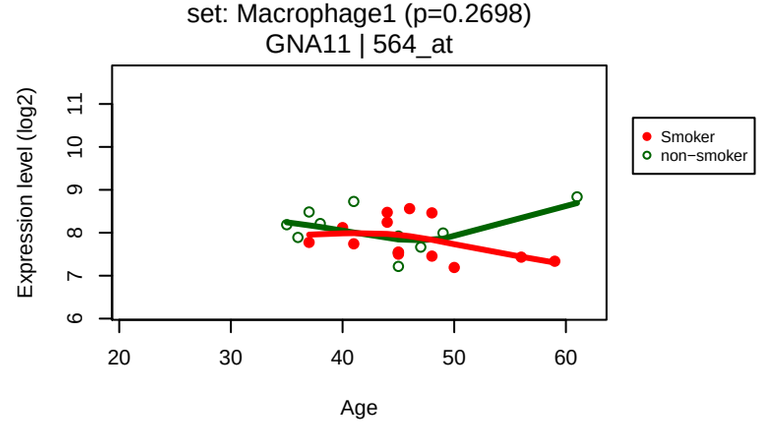

set: SmallAirway1 (p=0.9962)
GNA11 | 564_at
Expression level (log2)
Age
Smoker
non−smoker

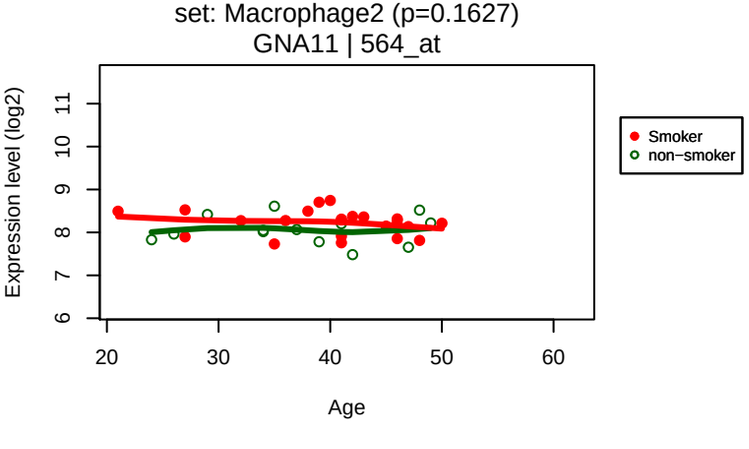

set: SmallAirway2 (p=0.0347)
GNA11 | 564_at
Expression level (log2)
Age
Smoker
non−smoker

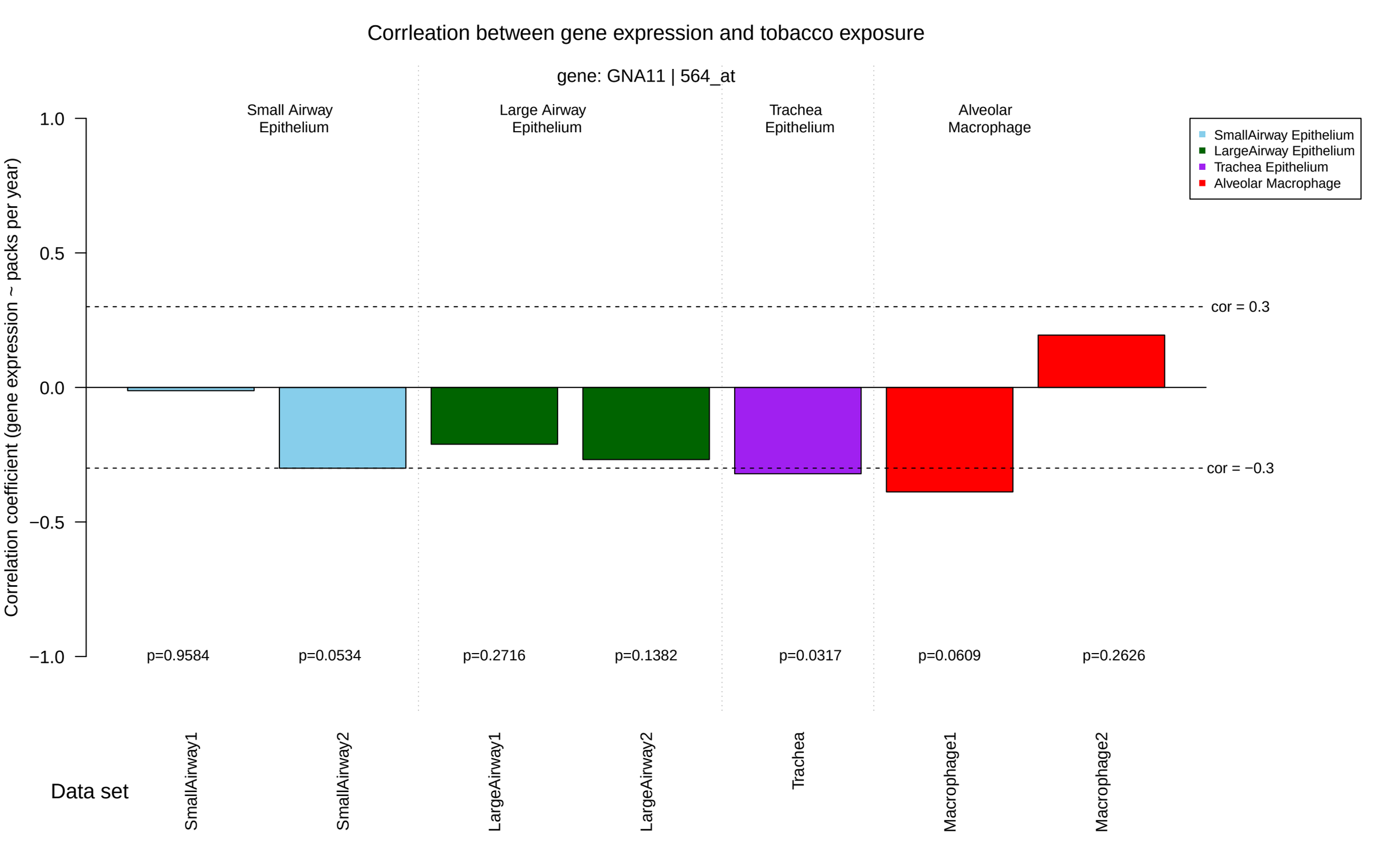

set: LargeAirway1 (p=0.1653)
GNA11 | 564_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5591)
GNA11 | 564_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0679)
GNA11 | 564_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2698)
GNA11 | 564_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1627)
GNA11 | 564_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNA11 | 564_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9584
p=0.0534
p=0.2716
p=0.1382
p=0.0317
p=0.0609
p=0.2626
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2